Parikshit Pareek

Ph.D. Student, Power System Analytics

Clean Energy Research Lab
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I am currently a Ph.D. student and Research Associate working on smart grid optimization and stability assessment methods. My goal is to develop interpretable and explainable machine learning models for supplementing our understanding of power system. I wish to find the solutions which will enable us to provide safe, affordable and green energy to the person who is at the bottom of the pyramid.

Education

2018-Present **Doctor of Philosophy**, *Nanyang Technological University*, Singapore.
School of Electrical and Electronic Engineering, NTU, Singapore

2016-2018 **Master's of Technology**, *Indian Institute of Technology Delhi*, India.
Centre for Energy Studies, IIT Delhi

2011-2015 **Bachelor's of Technology**, *Rajasthan Technical University Kota*, India.
Department of Electrical Engineering, RTU, Kota

Awards and Honours

Dr. Shankar Dayal Sharma (Former President of India) Gold Medal-2018, *IIT Delhi.*
Awarded for General Proficiency including Character & Conduct, Excellence in Academic Performance, Extra-Curricular Activities and Social Service among all M.Tech graduating students of IIT Delhi.

POSOCO Power System Award (PPSA)-2019, *POSOCO, India.*
Awarded for Master's Thesis on Power System by Power System Operation Corporation Limited (POSOCO), A Government of India Enterprise.

Prof. O.P Gupta Medal-2018, *IIT Delhi.*
Awarded for being the department rank one at Centre for Energy Studies, IIT Delhi.

Bhagirathi - Bashisht Tiwari Award-2018, *CES, IIT Delhi.*
Awarded for the highest CGPA among all graduating students of M.Tech Energy Studies IIT Delhi.

Shrimati Jawala Devi - Sita Ram Kaushik Award-2018, *CES, IIT Delhi.*
Awarded for best M.Tech Dissertation for the year 2018 at CES, IIT Delhi.

NTU Research Scholarship, *NTU Singapore.*
Awarded for outstanding graduate students for research leading to a Ph.D. at the University.

Ministry of Human Resource Development Scholarship-2016, *Government of India.*
Awarded for Two-year Master's degree based on GATE percentile.

Research Experience

2018-Present **Graduate Research Scholar**, *Clean Energy Research Lab, NTU Singapore.*

Research Title: Interpretable Machine Learning Models for Power System Application
- Developing ML models to understand power system's behaviour under uncertainty.
- Formulation of novel data-based inverse power flow approximation using Gauusian process.
- Defining ML model interpretability and supplementing the analytical understanding with it.

Research Title: A Novel Framework for Small-Signal Stability Constrained OPF.
- Development of sufficient condition for small-signal stability as Bilinear Matrix Inequality.
- Formulation of eigenvalue analysis independent SSSC-OPF based on convexification approach.
- Attacking the SSSC-OPF problem from sharp angles and providing new insights.

2016-2018 **Research Assistant**, *Electrical Energy Systems Lab, IIT Delhi, India.*

Research Title: Computationally Efficient Linear Optimal Power Flow for Power System Applications.

- Development of irregular polygon-based piecewise linearization of quadratic thermal limits.
- Development of renewable forecast based pricing and linear DR-OPF for distribution system.
- Post-Optimal Analysis based method for Solar PV and load uncertainty handling.

2013–2015 **Undergraduate Research Assistant**, *RTU Kota, India.*

Research Title: Condition Monitoring of Squirrel Cage Induction Motor using Stator Current Signal Monitoring

Publications

Parikshit Pareek and Hung D. Nguyen, "Gaussian Process Learning-based Probabilistic Optimal Power Flow" *IEEE Transactions on Power Systems*, Early Access, 2020

Parikshit Pareek and Hung D. Nguyen, "Probabilistic Robust Small-Signal Stability Framework using Gaussian Process Learning" *Electrical Power System Research*, Nov. 2020

Parikshit Pareek, Konstantin Turitsyn, Krishnamurthy Dvijotham, Hung D Nguyen, "A Sufficient Condition for Small-Signal Stability and Construction of Robust Stability Region" *IEEE PES General Meeting, August-2019*

Parikshit Pareek, and Hung D Nguyen, "Small-Signal Stability Constrained Optimal Power Flow: A Convexification Approach" *Preprint, ArXiv:1911.12001*

Parikshit Pareek, Chuan Wang and Hung D Nguyen, "Non-parametric Probabilistic Load Flow using Gaussian Process Learning" *Preprint, ArXiv:1911.03093*

Parikshit Pareek and Ashu Verma, "Piecewise Linearization of Quadratic Branch Flow Limits by Irregular Polygon" *IEEE Transactions on Power Systems*, Vol. 33, No. 6, pp. 7301-7304, 2018

Parikshit Pareek and Ashu Verma, "Computationally Efficient Day-Ahead OPF using Post-Optimal Analysis with Renewable and Load Uncertainties" ArXiv preprint arXiv:1804.06391, 2018

Parikshit Pareek and Ashu Verma, "Linear OPF with Linearization of Quadratic Branch Flow Limits" *IEEE IEEMA Engineer Infinite Conference*, Grater Noida, 2018.

Responsibilities

2020-2021 **Vice-President** External Committee, EEE Graduate Student Club, NTU Singapore

2019-2020 **Teaching Assistant** School of Electrical and Electronic Engineering, NTU Singapore

2019-2020 **Director** Outreach Internal Support Committee, EEE Graduate Student Club, NTU Singapore

2018-2019 **Member** EEE Graduate Student Club, NTU Singapore

2016-2018 **Teaching Assistant** Electrical Energy Systems Lab, CES, IIT Delhi, India

2016-2017 **Vice-President** of Energy Society, CES, IIT Delhi, India

References

Prof. Hung D. Nguyen, *Assistant Professor*, School of Electrical and Electronic Engineering, NTU Singapore, *hunghtd@ntu.edu.sg.*

Prof. Ashu Verma, *Associate Professor*, Centre for Energy Studies, IIT Delhi, India, *averma@ces.iitd.ac.in.*

Prof. S. R. Kapoor, *Professor*, Department of Electrical Engineering, RTU Kota, India, *srkapoor@rtu.ac.in.*